Traditional Rice Cultivation in Sri Lanka

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Content of the Lecture

1. Traditional Agriculture

Sri Lanka's farming systems particularly rice, other field crops and home gardening have evolved over thousands of years including a rich array of farming systems and the cultivated crops such as rice, grains, vegetables, fruits, spices etc. and livestock. New crop varieties emerged formally and informally. In addition, many farmers have selected local landraces. The long history of cultivation, presence of cultural diversity and wide range of ecological landscape situations present in the country have resulted in a wide variety of farming practices in Sri Lanka.

The excavations in citadel of Anuradhapura produced important evidence of iron technology, breeding of horses and cattle, and paddy cultivation, from cultural horizons nearly ten meters below the present ground surface. There is also incidental evidence (faunal, sedimentological) of water management associated with paddy cultivation. Agriculture would undoubtedly have been dominated by paddy, which can only be intensified in Sri Lanka's dry zone by the adoption of water management measures to control supplies from seasonal rainfall, streams, and perennial rivers (Deraniyagala, 2002)[1].

In the past the ingenious civilization widely known as 'Hydraulic Civilization of Sri Lanka' was supported by indigenous agricultural practices adapted to limiting factors such as uncertainty of rainfall, fragility of ecosystems, erosion of topsoil, damages from pest, diseases and wild animals etc. The communities lived in harmony with nature and were self-sufficient in food, autonomous in society and sustainable in agriculture. There are still some food materials, which are generally consumed by peasant sector remained with them. The arrival of Green Revolution in mid-20th century, which brought western agricultural technology (high input required varieties, soil devastating machinery, environmental hazardous agro-chemicals and inorganic fertilizer) has posed serious threat to the survival of the traditional agriculture system in Sri Lanka.

As once stated by Robert Chambers (1994)[2] that the indigenous knowledge has different modes of experimenting and learning. Methodologies and analytical tools adopted in modern science to accept or reject should be utilized with some modification in investigating the indigenous knowledge. Interpretation of any observation must be carefully made giving due consideration to the 'time tested facts'. However, inability to interpret any phenomenon does not mean that it is at myth. Many practices adopted in the traditional communities are blended with religious and spiritual beliefs and cosmic influence. Similarly, the Sri Lankan traditional agriculture has evolved with its unique features synchronized with socio-cultural and cosmo-spiritual dimensions into the bio-physical process found generally in western agriculture (Fig. 1). Thus, it is wise to use a 'package effect approach' rather looking for effects of each component separately. For example, the *nawa kekulama* is an improvement of the traditional kekulama considering the present environment, where it is to be practiced. Irrespective of the unknown and unseen influence of certain components, the null hypothesis can be tested for the whole package of *kekulama*.

[1] Deraniyagala SU. 2002. A note on man and water in Sri Lanka: from prehistory to history. In: Mendis DLO Water Heritage of Sri Lanka. Sri Lanka Pugwash Group, Colombo, p xix-xxi.

[2] Robert Chambers, 1994. Beyond Farmer First: Rural People's knowledge, agricultural research and extension practice. (Ed.) Ian Scoones and John Thompson. International Institute for Environment and Development

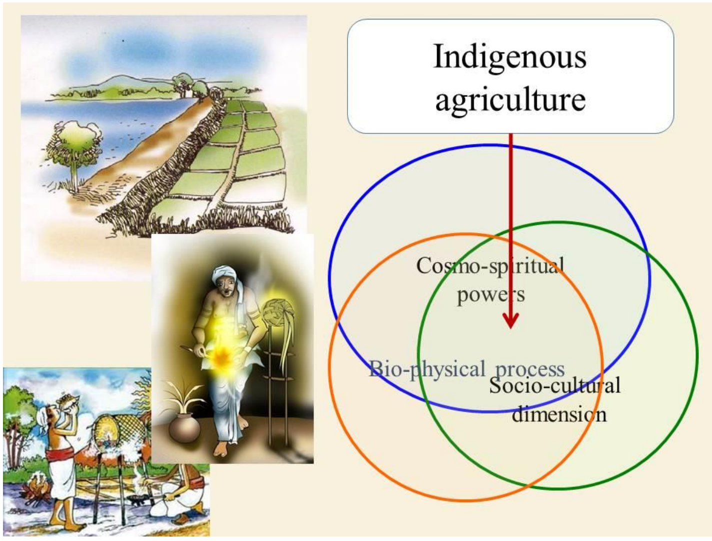

Fig. 1. Three dimensional indigenous agriculture

Knowledge on various practices and technologies has been built up over time. They could find best practices on how grains are stored for consumption as well as for seed requirement without any quality deterioration. Beliefs are synthesized with disasters, worries, failures and successes experienced through generations. They protected forest, watersheds, medicinal species and various pious places and materials. Simple tools and implements were developed to deal with many operations that the community had to perform in their daily life. Religious festivals were organized with different objectives but led to strengthen their social solidity and self-motivation. Materials had been identified from natural environment to be utilized for housing construction, basketry and other craft industries.

The diversity of the ecosystems provided a natural setting for the adaptation of agricultural system capable of producing foods to sustain large population. Traditional wisdom in agriculture and the living have not evolved within few decades. It is a long time-tested knowledge, which created an environmentally adapted, disaster tolerant and sustainable living system. Their agriculture had been adjusted to absorb any weather vagaries by shifting the cultivation time and selecting farming practices. They cultivated chena and paddy lands according to the seasonality of rains thus; at least they could get a successful harvest from cultivation.

'*Kekulama* (dry sowing), *Bethma* (shared cultivation), *Thaulu govithena* (tank bed cultivation) etc. are the best examples showing how they could avert the drought effects on their farming. Traditional communities made every attempt to conserve soil, water, and natural habitat. Food security was one of the in-built aspects of their culture. Use of groundwater for agriculture was never practiced by them, and it assured the water security. An adequate dead storage was found in tanks to be utilized during dry period for all purposes and had been the only source of water for cattle and wild animals. There was a broad diversity in flora and fauna and the availability of water in the tank during the dry period assured the survival of them. Sharing resources equally and the equity of ownership were the most striking features of their culture, which led to build up a peaceful and sustainable rural society. Environmental pollution was not a topic for discussion. With the disappearance of the features discussed above the whole system was subjected to deteriorate socially, physically and economically leaving vulnerability to disasters with them.

2. Traditional Rice Cultivation

As rice farming is not an overnight wonder, it has evolved while facing challenges imposed by nature in form of drought, flood, cyclone, epidemics etc. Thus, the skills of farming developed include various types of best practices, which could be adopted even within the present arena of rice cultivation (Fig. 2). Some of them are briefly described below:

Fig. 2. Traditional rice farming in Sri Lanka

According to documentary evidence, Sri Lanka had rice cultivation as early as 800 B.C. This is further reflected by the construction of massive irrigation structures, reservoirs, interconnected canals since 390 B.C., The Rice cultivation was not only an economic activity but a way of Life. Once, renowned as the granary of the east: offered more than 2000 indigenous rice varieties to the rest of the world. Rice cultivation in Sri Lanka used to be sacred and well thought out; the methods of production and the sanctity associated with the process of production made it a truly sustainable process. In this process, many of the traditional varieties of Sri Lankan rice known to contain higher amounts of glutamic acid, higher concentrations of vitamins, richer in fiber, a lower glycemic index and what was known to have nurtured a prosperous society were lost to the nation. At present, more than 95% of the paddy field is cultivated with the so-called semidwarf newly improved rice verities, which are harvested using chemicals, non-organic fertilizer and pesticides, which is a requirement for larger harvests hence the lower price. However, with the current trend of global awareness of the benefits of consuming organic food and the dangers of using chemical fertilizer and pesticides, traditional rice is gradually making a come-back. Farmers had many rice varieties developed using natural selection by observing adaptive ability of such varieties for their environmental factors such as tolerance to water situation, pest and disease prevalence, soil fertility etc. and for various social needs such as health, cultural functions, religious needs etc. Among them many varieties are found for specific purposes. Table 1 provides some of the important information of traditional rice varieties (Ceylon Digest, 2019)[3].

Table 1. Importance of Sri Lankan traditional rice varieties

Name of the rice variety	Importance
Suwandel	Exquisitely delicious white rice with an exquisite aroma. It is well known to promote fair and glowing skin; improve the functioning of the excretory system; improve vocal clarity; enhance male sexual potency and help control diabetes. It is also said to support a balanced growth of body. Its special milky taste makes it an ideal choice for festive occasions and ceremonies. Suwandel's nutrient composition consists of 90% carbohydrate, 7% crude protein, 0.7% crude fat,

[3] Ceylon Digest. 2019. Traditional rice varieties of Sri Lanka. Sri Lanka News Portal. Culture and Heritage.

	and 0.1% crude fibre. It is also known to contain higher amounts of glutamic acid and higher concentrations of vitamins than other more common rice varieties.
Kalu heenati	Its name literally means dark, fine grain. Kaluheenati is highly nutritious red rice with medicinal properties, perfect for daily consumption and is particularly recommended for lactating mothers. It is said to enhance physical strength and with its high fibre content it helps regulate bowel movement. It is effective in keeping diabetes under control as well as controlling the toxic effects of snake bites. The porridge 5 made from kalu heenati rice is recommended for hepatitis patients.
Ma-wee	Ma-wee is a reddish-brown rice variety with a unique texture that is low in carbohydrates, and rich in protein and fibre. It is also proven to have 25% to 30% lower Glycemic Index (GI) in comparison to other common rice varieties. It has a nutrient makeup of 84.5% carbohydrates, 9.4% protein, 3.6% fat and 1.1% fibre. Ma-wee was loved by the queens who believed that it helped maintain a slim, shapely figure. It is said to provide relief for burning sensation and cool the body. Ma wee rice consumed together with meat can reduce alcohol intoxication. It is recommended for tuberculosis patients and is an effective remedy for purging. It is also recommended for diabetes, tuberculosis, constipation, hemorrhoids and cardiovascular disease, and is known to controls corpulence. Ma wee rice is best when soaked prior to boiling. It is also revered for its historical importance in religious ceremonies. According to folklore Ma wee has been placed in caskets of sacred relics and the pinnacle (kotha) of dagabas.
Pachchaperumal	The word ‘Pachchaperumal’ means The Lord ‘Buddha’s colour’ and this rice variety has been considered as divine rice in traditional Sinhalese culture. It was often used in alms giving. This is a wholesome red rice variety, which when cooked takes on a deep rich burgundy colour. It is rich in nutrients and in proteins, and is an excellent choice for your everyday meal. Pachchaperumal is known to be a perfect diet for those with diabetes and cardiovascular disease.
Kuruluthuda	It is a delectable and nutritious red rice variety, which is rich in proteins and fibre. It has a pleasant taste. Kuruluthuda is said to improve bladder functioning, enhance male sexual potency and help evade Impotency.

Rathdel	This is a delicious red rice that provides relief to those suffering from cirrhosis. Porridge and soup made with rathdel can help fight against viral fever. It is recommended for rashes caused by mental stress and provides relief for ailments in the urinary system. It also helps flush toxic 6 excretory matters and cools the body. Roasted and ground rathdel raw rice tempered with ghee can be an effective remedy for purging. It is proven for preventing the formation of stones in the bladder and gall bladder. Porridge made out of rathdel rice, sarana (*Boerhavia diffusa*), sugar, raisins and fresh cow's milk is suitable for those suffering from tuberculosis and lung ailments.
Madathawalu	This is another traditional red rice variety that is highly recommended in Ayurvedic treatment to strengthen the immune system.
Hetadawee	Duration of the rice crop is 60 days, hence the name comes. It is a red rice variety, which helps control diabetes and provides relief for burning sensations and cools the body. It relieves ailments caused by biological imbalances; improves physical strength, and also an effective remedy for purging, blood vomiting and bleeding disorders.

Farmers from Hambantota District have reported that there are many traditional rice varieties tolerant to saline conditions of the soil. *Pachchaperumal, wanni dahanala, rathdel, dahanala, kuruluthuda and madathawalu* are among them (Dharmasena, 2007)[4]. In general, traditional varieties grow taller and susceptible to lodging. Farmers feel that under conditions prevailed in their fields and the management practices they could adopt traditional varieties could produce relatively higher yields compared to improved varieties. In order to get high yields from improved varieties they have to spend more money for inputs. Further, they mentioned that taste of traditional rice varieties is more than improved ones, and also boiled rice of traditional varieties can be kept for longer time without spoiling.

In the Registry of Rice Land Traces published by the Plant Genetic Resources Center, Peradeniya, there is more than one accession for many traditional varieties. This is mainly due to the wide variation of characters among these accessions of the same variety. This calls for an evaluation process of the varieties on their performance under the conditions it is cultivated.

[4] Dharmasena, P.B., 2007. Introduction of traditional rice farming for salinity affected areas in the Hambantota District of Sri Lanka. A Review report, Practical Action.

Strategies adopted in traditional cultivation for mitigating the impact of drought and flood are discussed below (Dharmasena, 2011)[5].

3. '*Bethma*' Cultivation

The bethma cultivation practice is adopted in poor rainfall seasons when the farmers cannot cultivate the entire paddy tract by using limited water in the tank. They gather and decide to redistribute temporarily the upper portion of the paddy tract mostly in equal size. This provides a part of their food requirement otherwise would end up with full abandonment of the paddy cultivation for that season. Through bethma, the limited tank water could be utilized efficiently without causing crop losses. Such a practice does not depend on the subsequent rainfall, which is not certain. Bethma can also now be practiced in combination with field rotation, and the farmers may decide to cultivate either paddy or other field crops. This decision usually depends on the water level in the tank. In some cases, the land distribution is proportional to the ownership share of land, but in most cases, it is found to be in equal portions. Allocation of the plots is usually done by either the Vel-vidane or the farmer organization.

4. '*Pangu Kariya'*

Farmers divide the maintenance works, such as, tank bund clearing, canal clearing, bund repair, sluice cleaning, etc. among themselves to prepare the irrigation system for the seasonal cultivation. This reduces the cost, creates sense of responsibility and ownership and accountability. Performance of works is in good quality and working together could strengthen the social cohesion. Nevertheless, regular maintenance, attending all repairs, ultimately could contribute to efficient water management and system sustainability.

5. '*Kekulama*' Cultivation

Dry sowing of paddy seeds early in the season is referred to as *kekulam govithena*. When dry sowing is done in tank upstream areas, it is called *vee hena* or *goda hena*. In the traditional kekulama method, the dry fields are ploughed with the country plough (*sinhala nagula*) to obtain a dispersed soil and burry the weeds. At the

[5] Dharmasena, P.B., 2011. Indigenous technology to Mitigate the risk of drought and flood. Practitioners' Guide Book on the Best Agricultural Practices for Drought and Floods in Sri Lanka. UNDP Sri Lanka. Chapter 8.

inception of rains, dry seeds are sown anticipating more rains soon. After sowing, the land is shallow ploughed to mix-up the seeds with soil. In some instances, when the field becomes adequately wet, the same *kekulama* could be practiced, but sprouted seeds need to be sown instead. In some paddy tracts and under some tanks, certain sections are sown to *kekulama* and then when the tank is full of water, other sections could be cultivated with normal wetland land preparation. A recent study (Dharmasena, 1989)[6] indicated that delaying cultivation without adopting *kekeluma* method would lead to high irrigation requirement failing to use a considerable portion effective seasonal rainfall for the cultivation (Fig. 3).

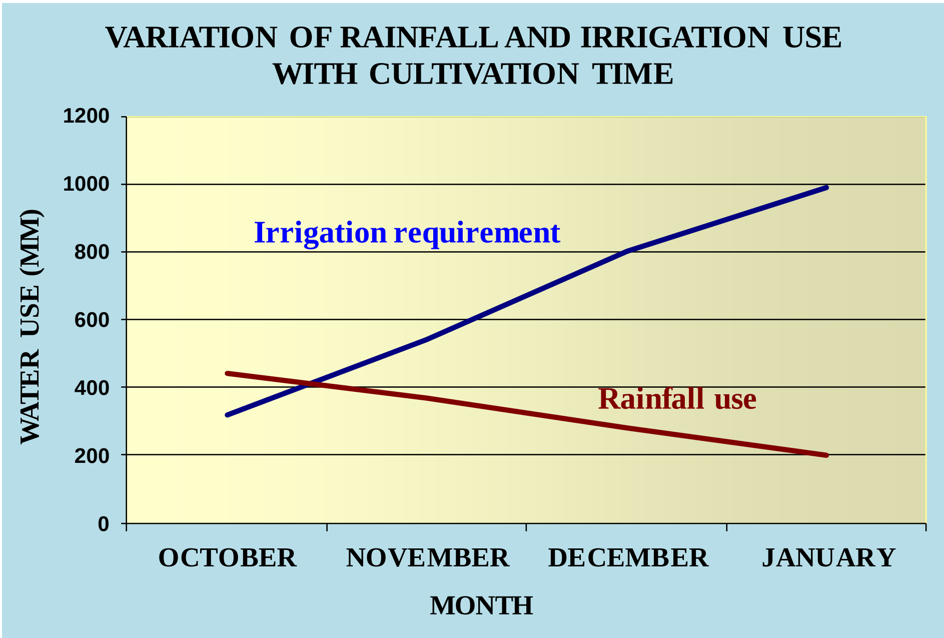

Fig. 3. Effect of delaying cultivation on irrigation requirement

6. ‘*Thavulu*’ Cultivation

When people realized that rainfall during coming *maha* season is very poor and tank does not get adequate storage, they decided to cultivate the upper part of the tank bed called *thavula* or *thavalla*. People fenced the *thavula* area and constructed a bund (*maha niyara*) along the lower boundary of the *thavula*. This area is somewhat flat

[6] Dharmasena, P.B., 1989. Optimum utilization of the storage in village tanks. Tropical Agriculturist. Dept. of Agriculture, Peradeniya, Sri Lanka 145:1-11.

and farmers cultivate this land expecting any rains during *maha* season to capture and use without allowing to flow into the tank bed. People believe 'would be rainfall' through traditional weather forecasting methods.

7. Crop Protection

In Sri Lanka, there were many traditional practices designed to protect crops from the damage caused by wild animals. During the last century, land use has changed considerably. In the wet zone forests, large scale tea and rubber plantations have sprouted. In the dry zones extensive irrigation projects have promoted paddy farming and colonization removing the forest. In this process poor farmers have been driven into more marginal lands and today these areas are often close to wild life reserves.

According to traditional practice, agricultural cultivation was a community activity. Cultivation started with the making of a vow to the gods to ensure the success of cultivation. Then cultivation was initiated by the village leader in one area at an auspicious time. These activities were followed by several agricultural practices: minimal tillage of the land; mixed cropping and seeding wherever possible; fencing activities at auspicious moments; crop protection like the cultivation of a small portion of land (kurulu paluwa) to attract birds for pest management in paddy, performing kems, a ritual or a religious rite, and if necessary supplement these actions with use of plants or plant extracts (bio-pesticides); Harvesting and heaping in the field for some weeks or months; threshing and separating a small portion for the Mangalya or other festival.

There are three categories of traditional practices to protect crops from wild animal damage. The first group is based on astrology, the second on the powers of the spirits and Gods, and the third involves the chanting of verses and the use of specific symbols. Often these different practices are combined. Upawansa (2000)[7] discussed these practices as below.

Astrological practices – Some astrologers are generalists, while others are specialized in health, agriculture and helping travelers. Astrology plays a dominant role in agriculture, especially in the cultivation of rice. Farmers believe that certain days are good for beginning cultivation. They also avoid certain days which

[7] Upawansa, G.K., 2000. Testing indigenous techniques to protect crops from wild animal damage. Compas Newsletter – July 2000.

they consider inauspicious or unlucky. Usually a Sunday is chosen to initiate work relating to paddy cultivation. The work is begun on an auspicious day at an auspicious time. Most farmers follow the astrological calendar or pancha suddiya to ensure success and avoid bad luck.

Spirits and Gods – If the people realize that the issue cannot be addressed by their strategies then they expect the support of Gods and spirit. One such example is that all farmers visit the temple and make offerings before they start cultivating their crop. Each season almost every farmer performs an offering and hangs a coconut in the field before cultivation begins. They also participate in the communal rituals held at the temple. After the harvest, farmers perform a ritual in the field before use. They believed that such practice could please the unseen forces. These activities are still taking place in some rural villages. Since ancient times rituals have been used in Sri Lankan agriculture to support crop growth and animal husbandry, and to chase away the wild animals or pests that damage the crops. The combination of spiritual practices, astrology and eco-friendly technologies have become customs. Despite the impact of the green revolution, many of these spiritual practices still exist, but in some instances, their full meaning is not fully understood by the younger farmers.

Pirith – *Pirith* is Buddha's teaching for laymen and involves chanting specific verses in a group. Each verse deals with some aspect of good living. Some of these prescriptions are used for crop protection. The verses are used to charm sand and water. These are then sprinkled thinly over the field. Chanting specific verses extracted from Buddhist teachings is done in a group. In some areas symbols are painted on an ola leaf and hung in the corners of the field. However, the performer is said to be having a pious life and he should refrain from robbery, sexual misbehavior, eating animal protein or drinking alcohol.

Manthra – The *manthra* is also chanting with specific sounds repeating the same version specific number of times. This causes a vibration in the environment. This influences the spirits to bring about the desired effect. In the mantra Gods or religious leaders, like Lord Buddha or the Prophet Mohammed, can also be called upon and their great achievements are recalled. Each mantra is different and depends on which animal is being addressed. When elephants are threatening the crops, the mantra must be accompanied by 12 placing a charmed coconut flower in the middle of the plot. If the animal concerned is a wild boar, a glowing fire stick is charmed and dipped in the paddy field.

To prevent rat attack sand and pebbles are taken from the field and these are then charmed. The sand is then sprinkled over the field while pebbles are buried in each of the corners. Charmed pebbles are also buried in each corner of the field to ward off monkeys. Birds are kept away by burying charmed mustard seeds and sand in the centre of the field.

Yanthra – A symbolic drawing preferred by a particular spirit is hung or kept in a specific place expecting the blessings of unseen power to carry out their activities or to live without any threats. Drawing a *yantra* involves, following certain laws. If these laws are not carefully followed not only will the *yantra* have no effect, but evil things may happen. For the spirit to occupy the *yantra* it has to be enlivened with specific verses, or mantras. In agriculture, the use of *yantra* is widespread. Generally a *yantra* is placed in the centre of the rice threshing floor. An abstract geometrical drawing is used: three concentric circles and eight radial lines with different drawings on the outside: The *yantra* is placed or drawn on the threshing floor, certain items such as an oyster shell, a coconut, a piece of iron are placed on it, together with a few bundles of paddy.

Kem karma – The practice of *kem* is very widespread in rural Sri Lanka. A *kem* is a kind of practice, technique or custom that is followed in order to obtain some favourable effect such as relief from a specific illness. For example, washing in a pool of water immediately after a crow washes in that pool is believed to bring relief to people suffering from certain infirmities. A requirement in this *kem* is that the patient should wash without speaking or making much noise.

Some *kem* combine the use of astrology with the use of certain plants or herbs. Other *kem* depend on the use of specific plants and mantras. These traditional practices have survived because they must be effective. If these had no real effect, they would have disappeared long ago.

There are also *kem* that do not involve any belief in spiritual beings or gods. These *kem* are based on a careful observation of nature and natural phenomena. Some *kem* are mechanical methods, like the lighting of fire torches. These torches are made 13 using a piece of saffron robe for the wick and sticks of trees *wara* (*Calatropis gigantea*), *kaduru* (*Pagiantha dichotoma*) or *gurula* (Leea indica) for the handle. There are various conditions that have to be met to make the working of *kem* successful. With some *kem*, women are prohibited from entering the field altogether, while other kem have to be performed by women only or even by pregnant women only. The effectiveness of a *kem* can be nullified if the person is exposed to a *killa* or impurity caused by eating certain food (especially meat).

8. Adaptable Practices

The paddy sector is extremely vulnerable to climate change impacts. Over the years, on many occasions, farm lands were heavily damaged due to floods and droughts. Traditional agriculture practices coupled with local paddy varieties have proven to be more successful in facing climate change events such as droughts and floods.

There are only a handful of traditional paddy varieties in existence today in Sri Lanka. These are: *Suwandel, Rathdel, Kaluheenati, Ma-Wee, Kuruluthuda, Pachchaperumal, Madathawalu, Hetadha Wee, Hondarawalu, Girisa, and Heenati.* These traditional paddy varieties have strong characteristics that help them survive climate change impacts such as droughts, heavy rains, and floods, compared to newer varieties used in chemical intensive paddy cultivation. This vigour is based on certain characteristics unique to traditional paddy varieties.

The Government has clearly identified the problem of farmer and country's agriculture. Rehabilitation of irrigation schemes at all scales and development of new schemes all over the country are the main steps in planning to improve the agriculture in the country. In this context paddy cultivation will be still encouraged to face the challenge of achieving the national annual production requirement of 3.5 million metric tons of rice by the year 2025. Today one of the major problems in paddy farming is the deterioration of resource base causing gradual decline in the productivity and profitability of paddy farming sector.

In order to address the above issues, following practices are recommended (Dharmasena, 2014)[8].

- Ways of reducing the cost of production especially by addressing the problems

[8] Dharmasena, P.B., 2014. Traditional Rice Farming: Back to practice. Technical Report. ResearchGate. Available at: https://www.researchgate.net/publication/355186545_TRADITIONAL_RICE_FARMING_BACK_TO_PRACTICE

of weeds, pest and diseases.

- Adoption of low cost practices to enhance soil fertility such as use of compost, green manure, cow dung, poultry manure, liquid fertilizer etc.
- Methods to increase profitability of rice farming by lowering the cost of input use and offering high market price for traditional rice varieties.

As mentioned at the beginning traditional rice farming includes many practices during the cropping period. However, it may be difficult to introduce all at once as these practices are now new to them. Considering this fact, following four main components can be included as to make the package of traditional rice farming.

i. Cultivation of traditional rice varieties
ii. Land preparation and water management taking cognizance from traditional farming
iii. Applications of fertilizers of organic origin (straw, green manure, cow dung, poultry manure, liquid fertilizer etc.)
iv. Management of weeds through hand weeding, mechanical weeding and water management
v. Management of pest and diseases by practicing *kem* krama (rituals), maintaining biodiversity and using bio-pesticides

Present health problems they face with the use of agro-chemicals and inorganic fertilizer can make them to think that the traditional rice farming is only the possible alternative solution. As discussed early they can use various traditional rice varieties available around them such as *kuruluthuda, pachchaperumal, suwandel, rathkanda, kalu heenati* etc. (Dharmasena, 2014).

Land preparation - If the 'Sinhala nagula and buffalo' practice is not possible, farmers can use two wheel tractor with the rotary to plough the field. Partially burnt paddy husk and *gliricidia* leaves can be spread over the field two weeks after first ploughing. Bunds are repaired before leveling the field.

Planting method – Broadcast sowing is preferable. Seed rate of 45 – 50 kg/acre with traditional varieties is sufficient. They use a rate of 60 – 70 kg/acre with improved varieties.

Water management – During a month time, land preparation and sowing have to be completed with irrigation water. Subsequently, irrigation water is supplied once in 4 days.

Addition of organic matter – Farmers can apply partially burnt paddy husk, cow dung, liquid fertilizer, *gliricidia*, g*ansooriya*, *pawatta* (*adathoda*) and rice straw.

Pest management – Spreading ash over the field, walking across the field, *kem krama*, placing crushed *kalawel* packed in bags, astrological methods, bio-pesticides etc.

Farmers believe that use of inorganic fertilizer does not increase yield much in saline soils, but a satisfactory yield could be obtained with organic fertilizer with low or no cost. They could not observe much difference in weed population between fields cultivated with traditional varieties and improved varieties. Crop growth at the beginning is observed slow in traditional varieties, but shows a rapid growth rate at development stage. Yields could be obtained as 90 – 100 bushels per acre irrespective of the variety (Dharmasena, 2007)[9].

9. Performance Evaluation

It was found much difficult to evaluate the overall performance of the farmers and farming practices due to the wide variability of inputs, practices, effects and impacts. Further, the challenge in evaluating some aspects such as social cohesion, mental satisfaction etc. remains unresolved in economic evaluations. Therefore, the author has developed a performance index embedded with quantitative as well as qualitative aspects of the evaluation process (Dharmasena, 2010)[10].

A matrix was prepared with practices (use of traditional rice varieties, organic manure, medicinal liquids, herbal seed treatments, astrology or *neketh*, *manthra*, *yanthra*, rituals and *kem,* non-use of artificial fertilizer, insecticides, and weedicides and adoption of appropriate spacing) against their performance in relation to economic benefits, low labour use, low production cost, improved family health/ nutrition conditions, environmental benefits, increased yield, mental satisfaction and social cohesion.

10. A Comparison Between Traditional and Modern Rice Farming

An assessment was made in Moneragala District to compare the traditional rice

[9] Dharmasena, P.B., 2007. Introduction of traditional rice farming for salinity affected areas in the Hambantota District of Sri Lanka. A Review report, Practical Action.

[10] Dharmasena, P.B. 2010. Essential components of traditional village tank systems. In: Proceedings of the National Conference on Cascade Irrigation Systems for Rural Sustainability. Central Environmental Authority, Sri Lanka, 9 December 2010.

farming and modern high input based rice cultivation through a comprehensive field study carried out in 2007/08 *maha* season and 2008 *yala* season. The study included soil analysis, crop growth measurements and field surveys for crop economics and pest and diseases incidence. The summary of findings (Dharmasena, 2010)[11] is given below.

Soil Status:

- In any soil if EC value is larger than 2 dS/m, then the field is considered as salt affected. If the value exceeds 5 dS/m, then the soil is considered not suitable even for paddy cultivation. Few varieties such as At 354, At 401, Bw 400 etc. and traditional varieties such as *Pokkali, Nonabokra, Muppangam* etc. are tolerant to a salinity level (EC) up to 5 dS/m.

After three months with traditional cultivation practices, the salinity level could be brought down to a level lower than 5 dS/m.

- There is no significant difference observed between soil pH values of rice fields cultivated with modern and traditional practices. However, a slight increase in pH of traditional rice fields and slight decrease in pH of conventional rice fields could be observed after three months of cultivation.
- Soils are low in available P. Available soil P had been higher in traditional fields than that in modern farming fields at the beginning, but an increase of available P in soils after three months of cultivation was observed as 19 % and 10 % in traditional and modern cultivation fields respectively.
- Exchangeable K in traditional farming sites was relatively high even at the beginning. However, K had not been a problem for rice cultivation. The soil K has been increased by 52 % and 33 % in traditional and modern farming fields respectively after three months of cultivation.
- Soil organic matter content in all rice fields studied was adequate to keep the soil condition favourable for rice cultivation. However, in traditional farming sites an average increase of 8 % could be observed during a three month cultivation period, while an average drop of 10 % was reported from modern farming sites.
- Although the rice plant can grow in heavy soils, it also needs some form of soil aeration, which facilitates the root respiration process. In the absence

[11] Dharmasena, P.B., 2010. Assessment of Traditional Rice Farming: A Case Study from Moneragala District of Sri Lanka, COMPAS Project, Future In Our Hands Development Fund Badulla 2010 Indigenous Knowledge Practices Promotion Programme.

of a structure, soil is more compacted and it should be improved with addition of organic matter to become physically fertile for paddy farming. Results indicated that soils with traditional farming were 46 % less compacted compared to that with modern farming.

11. Growth Performance of Rice Crop:

- Basic plant characters such as plant height, leaf length and width do not differ much due to variety and location.
- Tiller density was high in modern farming fields most probably due to high number of plants established per hill. Average number of seeds found in a tiller was about 80. A clear reduction in tiller density was observed in *yala* season compared to previous *maha* season irrespective of varieties and locations due to water shortage.
- Tiller productivity of rice plant has seriously dropped in *yala* season mostly due to shortage of water. It decreased from the average value of 80 observed in maha 2007/08 season to 32 in yala 2008 season.
- Grain weight of traditional varieties is 11 % higher than that of improved varieties. However, 1000 grain weight has decreased from 25 g. observed in *maha* season to 17 g in *yala* season.
- Both traditional and improved rice varieties performed equal grain filling percentage (80 %) in the rice fields studied.
- The average leaf area was relatively low in traditional varieties when compared to improved varieties. This is a disadvantage of traditional varieties in photosynthesis.

12. Crop Economics:

- The yield variation within a geographical area attributes to reasons such as use of different amounts of inorganic fertilizer, response to pest and disease incidence through various agro-chemicals, inheritance of soil properties and irrigation water management etc. Analysis showed that the relationship between input cost and the yield is linear and shown by the following equation.

  Yield (kg/ha) = 0.034 . Input cost (Rs./ha) – 95, $r^2 = 0.62$*
- Cost of production and the return to investment do not vary much according to the input levels.
- Farmers should be encouraged to move towards optimum input level for achieving a satisfactory income from paddy farming.

- There is a need to find ways and means to reduce the present input cost.
- Marketing network should be established to enable farmers to sell their produce at a higher price.
- Machinery cost for traditional farming was relatively low. Modern farmers spent 15 % more on tractors. Of total investment made during *maha* 2007/08 season, farmers spent 42 % and 28 % for machinery use in traditional and modern farming respectively.
- Material cost includes cost of fertilizer, agro-chemicals, seeds and others. In traditional farming cost for application of organic fertilizer sources and bio-pesticides was included in the material cost. Material cost could be curtailed by 55 % with adoption of traditional farming.
- Total labour use for cultivation was in the range of 60,000 – 80,000 Rs./ha, and it was in the range of 60-75 % of the total input cost. Use of family labour in traditional farming was found relatively high compared to that in modern farming. As a result expenses to be incurred for hiring labour in traditional cultivation could be reduced by 57 %.
- The average total investment made to produce one kg of paddy from traditional farming was only about 12 Rs. while it was about 17 Rs. from modern farming.

13. Resources Productivity: (Dharmasena, 2010[12])

- Due to low yields obtained from traditional farming average land productivity was Rs. 39,800/ha, and it was Rs. 47,700/ha from modern farming. Thus, land productivity was 17 % low in traditional farming.
- Farmers expect a seed productivity of 50 under normal condition. However, the observed average seed productivities were below this value and in the range of 30-40.
- Labour productivity in traditional farming was 32 % higher than that in modern farming. Further, in traditional farming farmers can earn more than Rs. 1,000 per day, which is a reasonable rate under present circumstances.
- Capital productivity in traditional farming is 33 % higher when compared to improved farming. Family labour can be used more effectively in

[12] Dharmasena, P.B., 2010. Assessment of Traditional Rice Farming: A Case Study from Moneragala District of Sri Lanka, COMPAS Project, Future In Our Hands Development Fund Badulla 2010 Indigenous Knowledge Practices Promotion Programme.

traditional farming.

- When family labour was not considered in the calculation it was found that in traditional farming, production of one kg of paddy could make a net return of Rs. 20.50, while it was only Rs. 15.70 from modern farming.

14. Crop Protection: (Dharmasena, 2010[13])

- Average pest: predator ratio in modern farming fields was 0.83 and below the required level of 2.0 for natural protection. The value with traditional farming was 2.4 indicating that the crop would be looked after well by the natural processes.
- Weed biomass remaining after harvest will be an important organic matter source for the next season. An average amount of 11 mt/ha (fresh weight) has remained on land after removal of paddy grains with straw during harvesting. This amount does not depend on the two farming practices. Variation observed may attribute to soil fertility level, water supply and the farmers' management practices.
- Increased biodiversity is the factor behind the successful protection of rice crop. In addition to rice plant about 50 plant species were identified in traditional farming fields, while it was 27 found in modern rice farming fields.
- Number of broad leave species found in traditional farming field was 23, while only 10 species found from modern rice fields. This is a consequence of applying weedicides, which destroyed all weeds including broad leaves, which are desirably contributing to the growth of rice plant.

15. Overall Performance Index (OPI): (Dharmasena, 2010)

- The OPI was developed to assess the performance of practices on various types of benefits anticipated by the farmer on economy, environment, family health, mental satisfaction etc.
- Advantages of the OPI are: it can assess the adoption level of different farmers; it determines effectiveness of different practices; it shows relative performance by farmers; it can be used to monitor the progress of farmers; it can be used for evaluating performance before the project terminates; and farmers can be guided to improve their practices.

[13] Dharmasena, P.B., 2010. Assessment of Traditional Rice Farming: A Case Study from Moneragala District of Sri Lanka, COMPAS Project, Future In Our Hands Development Fund Badulla 2010 Indigenous Knowledge Practices Promotion Programme.

- The OPI matrix includes a set of practices (traditional varieties, organic manure, liquid fertilizer, non-adoption of chemical inputs and practice of *yanthra*, *manthra*, *kem*, *puja* etc.) and their expected performance (economic, labour use, cost of production, family health/ nutrition, environment, yield, mental satisfaction and social cohesion).
- The OPI matrix indicates following facts in traditional farming.
  - Most effective practices in traditional farming are: use of traditional varieties; non-use of inorganic fertilizer; and use of organic manures. Other important applications are astrological practices, non-use of weedeicides and use of *kem karma*.
  - People gain mental satisfaction mostly from use of traditional varieties, use of rituals and non-use of inorganic fertilizer, insecticides and weedicides.
  - Benefits to the environment mostly come from the practices such as non-use of insecticides, use of organic manures and non-use of inorganic fertilizer.
  - Family health of the traditional farm families can be assured with use of traditional rice varieties, use of organic manures and liquid fertilizer and non-use of insecticides and weedicides.
  - Labour use would become minimal if the farmers do not use insecticides and inorganic fertilizers and apply astrological practices, *yanthra* and rituals.
  - Cost of cultivation can be curtailed by a considerable amount if they use organic manure instead of inorganic fertilizer, use of *kem* and astrological practices instead of using insecticides and adopt systematic planting practices.
  - Social relationships in the community will be improved if the farmers use traditional rice varieties, follow astrological practices, adopt *kem karma* and adhere to systematic planting.
  - Economic benefits could mostly be gained from the use of traditional varieties and organic manures and non-use of inorganic fertilizer, insecticides and weedicides.